

# Evaluating the Impact of Discovered Causal Structures on Structure-Aware Shapley Methods: An Experimental Pipeline

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**Abstract.** Structure-aware Shapley methods promise more faithful feature attributions by embedding causal graph topology into the credit-assignment process. In practice the causal graph is rarely known: practitioners must infer it from data using automated discovery algorithms whose outputs are noisy and error-prone. This paper introduces an experimental pipeline that feeds automatically discovered graphs (PC and DirectLiNGAM) into three structure-aware methods, Asymmetric Shapley (ASV), Causal Shapley (CSV), and Shapley Flow, and measures attribution quality degradation along two dimensions: Magnitude Divergence ( $\Delta M$ ) and Sign Disagreement ( $D$ ), each against three reference configurations. Experiments on a 50-feature confounded synthetic dataset and the Sachs cell-signaling benchmark show that ASV is the most stable under graph errors (both  $\Delta M_{\text{oracle}}$  and  $D_{\text{oracle}}$  are the lowest divergence in all comparisons), while CSV and Flow amplify discovery errors into large attribution shifts. The pipeline also provides a dual-signal diagnostic, based on joint  $\Delta M_{\text{base}}$  and  $\Delta M_{\text{disc}}$ , standardized metrics, that identifies structurally unreliable nodes without needing a ground-truth graph, and adapt to any dataset and model output dimensions.

## 1 Introduction

Explainable AI has moved from a research preference to a deployment requirement across critical decision-making domains such as credit scoring, healthcare, and public resource allocation, where model decisions must be justified to affected individuals and regulatory bodies [1]. Shapley values have emerged as a gold standard for local feature attribution, but traditional implementations assume feature independence, which in practice means treating all permutations as equally probable during coalition formation, or just sampling marginal distributions when they are correlated, this could give a false credit when the true generative process involves causal dependencies among inputs [4, 5].

Recent literature has introduced structure-aware Shapley frameworks that embed causal graph topology into the attribution process or modify the sampling strategy to capture dependencies. Asymmetric Shapley Values (ASV) restrict permutations to causal orderings [2]; Causal Shapley Values (CSV) replace observational marginalization with Pearl’s do-calculus [4]; and Shapley Flow reframes credit assignment from nodes to directed edges [11]. All three promise attributions more faithful to the underlying causal mechanism, but every one of these methods treats the causal graph as given.

In practice the graph is almost never directly observ-

able: practitioners must rely on automated discovery algorithms such as the PC algorithm [10] and DirectLiNGAM [9] to infer it from observational data. These estimated graphs are inherently noisy, subject to false-positive and false-negative edge errors, and sensitive to assumption violations [3].

We build an experimental pipeline that feeds automatically discovered graphs into the three methods and measures, for the first time, how much attribution quality degrades when the graph is imperfect. Evaluation is performed on a controlled synthetic dataset (50 features, known linear causal structure) and the real-world Sachs cell-signaling dataset [7], where an established consensus DAG allows external validation.

To answer the central question consistently we track two dimensions: the *magnitude* of the attribution assigned to a feature and the *sign* (direction) of that attribution. A graph that merely rescales attributions preserves the qualitative story told to an affected individual; a graph that inverts signs tells the opposite story. Each dimension is operationalized by a single metric used identically across three comparison configurations (vs. baseline, vs. oracle, and cross-discovery), establishing a universal diagnostic framework.

## 2 Conceptual Framework

### 2.1 Explaining Models, Not Nature

One distinction matters before going any further, the goal here is not to fully recover the physical laws of the system the model was trained on, but to capture enough of its causal structure to make the attribution computation more honest [1]. A practitioner does not need a complete causal model of the world; they need a plausible graph of how the input features relate to each other so that structure-aware methods can encode those relationships as constraints, rather than treating every feature as an independent, interchangeable partner. Physical interventions are not required and often not possible; what matters is that the graph reflects the real dependencies well enough to guide the credit-assignment process in the right direction.

### 2.2 Structural Causal Models and Credit Attribution

A Structural Causal Model (SCM) [6] describes a system as a set of equations  $x_i = f_i(\text{Pa}(x_i), \varepsilon_i)$  where each variable is determined by its causal parents and independent noise. When a full SCM is available, credit attribution becomes more precise: you can intervene on a variable, fixing it to a specific value while cutting its incoming edges, and trace how that change propagates through the graph to the final prediction. This is the intuition behind CSV and Shapley Flow, both of which are designed around this causal intervention idea. CSV approximates the post-interventional distribution using Pearl’s do-operator for each coalition evaluation; Flow routes credit along the actual graph edges so that structural paths determine how much each node contributes. In practice, a fully specified SCM is almost never available. The implementations here do not reconstruct structural equations: CSV uses a conditional Gaussian approximation along discovered parent sets, and Flow uses a binary foreground/background activation rule. Both methods still benefit from the graph topology, but the credit assignment is an approximation of the full interventional ideal rather than an exact application of it.

### 2.3 Causal Discovery

Two prominent discovery algorithms accessible to any practitioner are used in this pipeline. Both algorithms operate on the  $X$ -only training partition.

**PC [10].** A constraint-based method that recovers the causal skeleton through conditional independence tests (Fisher-z,  $\alpha = 0.05$ ) and applies Meek’s orientation rules. Undirected edges are resolved by orienting them from lower to higher feature index.

**DirectLiNGAM [9].** A functional causal model that exploits linear non-Gaussian structures to simultaneously identify causal ordering and structural coefficients. Edges with  $|\text{coefficient}| < 0.10$  are pruned.

### 2.4 Traditional Shapley Values

Shapley values originate in cooperative game theory as the unique fair allocation of total payoff among players [8]. The marginal contribution of feature  $i$  to coalition  $S$  is  $v(S \cup \{i\}) - v(S)$ , where  $v(S) = \mathbb{E}[f(X) \mid X_S = x_S]$ . Averaging over all orderings gives:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \quad (1)$$

The formula is the unique solution satisfying four axioms: *Efficiency* (attributions sum to the total model output minus the baseline); *Null Player* (a feature that never changes the coalition value receives zero credit); *Linearity* (attributions for a weighted sum of games equal the weighted sum of their attributions); and *Symmetry* (two features with identical marginal contributions to every coalition receive equal attribution). Together these axioms guarantee a principled, unambiguous allocation: no matter how complex the model, the credit is divided exhaustively, no player is penalized for doing nothing, and identical contributions earn identical rewards. Symmetry is especially important for its fairness interpretation, it prevents arbitrary tie-breaking and ensures that attribution differences reflect genuine differences in predictive contribution rather than the order in which features were considered.

Where Symmetry becomes problematic is when the input features are not, in fact, interchangeable. If  $X_i$  is a causal parent of  $X_j$ , treating them as symmetric partners means the uniform average over orderings includes orderings where  $X_j$  appears before  $X_i$ , orderings that reverse the real generative sequence. These orderings let  $X_j$  absorb credit that, causally, belongs to  $X_i$ , because in the true data-generating process  $X_j$ 's values are partly determined by  $X_i$ . The result is a systematic misattribution from upstream causes to downstream effects whenever features are correlated through causal structure.

## 2.5 Structure-Aware Methods

All three methods preserve Efficiency, Linearity, and Null Player while modifying Symmetry to accommodate causal topologies.

**Asymmetric Shapley (ASV).** Replaces the uniform distribution over orderings with a distribution placing mass only on topological orderings of the feature DAG [2]:

$$\phi_i(\text{ASV}) = \mathbb{E}_{\pi \sim U(\Pi_{\text{topo}})} [v(P_i^\pi \cup \{i\}) - v(P_i^\pi)]. \quad (2)$$

The coalition value remains observational; the graph enters only through the permutation set  $\Pi_{\text{topo}}$ .

**Causal Shapley (CSV).** Keeps the symmetric formulation but replaces the observational value function with an interventional one based on Pearl's do-operator [4]:

$$v_{\text{do}}(S) = \mathbb{E}[f(X) \mid \text{do}(X_S = x_S)]. \quad (3)$$

CSV uses  $v_{\text{do}}$  in the standard Shapley weighting formula. The interventional value is estimated by post-interventional sampling along the graph-defined parent sets using a conditional Gaussian approximation (Appendix B).

**Shapley Flow.** Moves from node-based to edge-based attribution [11]. The players are the directed

graph edges:

$$\phi_{u \rightarrow v} = \mathbb{E}_{\pi \sim U(\Pi_E)} [V(E_{u \rightarrow v}^\pi \cup \{u \rightarrow v\}) - V(E_{u \rightarrow v}^\pi)], \quad (4)$$

and node importance is  $\phi_u = \sum_{v: u \rightarrow v \in E} \phi_{u \rightarrow v}$ . Each node takes its foreground value if it has any active incident edge; otherwise its background value. With no graph there is no game to play.

## 3 Experimental Setup

Figure 1 shows the end-to-end experimental pipeline. Two datasets are evaluated across three structure-aware Shapley methods under three causal graph sources; the resulting attributions are measured along magnitude and sign dimensions.

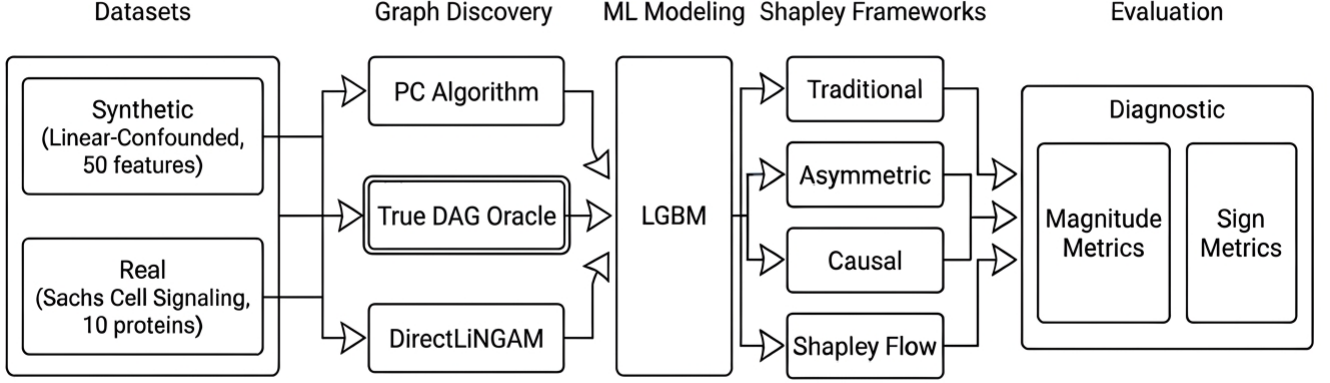
### 3.1 Datasets

**Synthetic (Linear-Confounder).** Strictly linear structural equations  $X_j = \sum_{i \in \text{Pa}(j)} w_{ij} X_i + \varepsilon_j$  ( $\varepsilon_j \sim \mathcal{N}(0, 0.5)$ , coefficients  $\in \text{Uniform}(0.5, 2.0)$ ). Erdős-Rényi graph ( $p = 0.07$ ,  $87 X \rightarrow X$  edges, 50 features, 15 direct parents of  $Y$ ). Five hidden confounders each influence 2–3 features, creating a noisy scenario for both discovery algorithms. 1000 observations.

**Sachs Cell Signaling [7].** 7,466 simultaneous measurements of 11 protein concentrations in stimulated human T-cells. Akt (akt) is the regression target; 10 proteins are input features. A consensus reference DAG (17  $X \rightarrow X$  edges, 3 direct  $X \rightarrow Y$  edges) provides oracle ground truth. Preprocessing: log1p transformation, then a Standard Scaler fit on the training partition only. Both datasets use an 80/20 split, seed 42.

### 3.2 Predictive Model

LightGBM regressor fitted with Optuna hyperparameter optimization (50 TPE trials, RMSE objective). Feature selection is disabled so all features participate in the Shapley computation. Table 1 reports performance;  $\hat{\sigma}$  is the standard deviation of model predictions over the test set and normalizes all magnitude metrics; the normalization rationale is explained



**Figure 1: Experimental pipeline overview.** End-to-end flow from data generation and causal discovery through Shapley computation to evaluation, showing how the two datasets, three graph sources, and three structure-aware methods feed into the magnitude and sign metrics.

in Section 3.4.

**Table 1: Model performance.**  $\hat{\sigma}$  normalizes all  $\Delta M$  metrics.

| Dataset   | R <sup>2</sup> | RMSE  | MAE   | $\hat{\sigma}$ |
|-----------|----------------|-------|-------|----------------|
| Synthetic | 0.900          | 1.72  | 1.36  | 4.71           |
| Sachs     | 0.679          | 88.78 | 17.82 | 116.76         |

### 3.3 Discovery and Computation Settings

Both discovery algorithms are configured as described in Section 2. Because neither algorithm produces  $X \rightarrow Y$  edges, an augmentation step appends direct  $X_i \rightarrow Y$  edges for every feature, uniformly applied to the True DAG as well, guaranteeing Shapley Flow has a route to accumulate credit for every node.

All Shapley methods use Monte Carlo approximation with  $T = 100$  permutations. Background reference: 30% of training rows (240 synthetic; 1,791 Sachs). Up to 100 test instances explained. CSV uses  $M = 10$  inner interventional samples per coalition. Table 2 summarizes what each method changes relative to Traditional Shapley.

**Table 2: What each method changes relative to Traditional Shapley.**

| Prop.     | Trad.     | ASV        | CSV         | Flow         |
|-----------|-----------|------------|-------------|--------------|
| Players   | Features  | Features   | Features    | Edges        |
| Perms.    | $N!$ ord. | Restricted | Restricted  | $ E +1$      |
| Value fn. | Obs.      | Obs.       | Intervent.  | Fg/bg act.   |
| Graph     | None      | Ordering   | Order+cond. | Players role |

### 3.4 Evaluation Metrics

Every comparison asks two questions about a (method, graph) result against a reference: did the attribution change in **magnitude**, and did it change in **direction**? Let  $\phi_{i,f}^{m,G}$  be the Shapley value for feature  $f$ , instance  $i$ , method  $m$ , graph  $G$ .

**Magnitude Divergence ( $\Delta M$ ).** At instance level, the signed gap between attribution magnitudes is  $\delta_{i,f}^{\text{ref}} = |\phi_{i,f}^{m,G}| - |\phi_{i,f}^{\text{ref}}|$ . Squaring and averaging over all explained instances  $\mathcal{I}$  gives a feature-level root-mean-square divergence, then divided by  $\hat{\sigma}$ :

$$\Delta M_f^{\text{ref}} = \frac{1}{\hat{\sigma}} \sqrt{\frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} (\delta_{i,f}^{\text{ref}})^2}. \quad (5)$$

The global dataset-level score averages across all  $N$  features:  $\Delta M^{\text{ref}} = \frac{1}{N} \sum_f \Delta M_f^{\text{ref}}$ .

The denominator  $\hat{\sigma}$  is the standard deviation of model predictions over the test set (reported in Table 1). Dividing by  $\hat{\sigma}$  expresses every divergence as a fraction of the model’s own output spread, making the numbers directly comparable across datasets, models, and output units. A value of  $\Delta M = 0.10$  means the typical attribution shift for that feature equals 10% of the model’s prediction spread, regardless of whether the model outputs loan amounts in thousands of dollars or protein concentrations in arbitrary units. Without this normalization, the synthetic and Sachs numbers ( $\hat{\sigma} = 4.71$  vs. 116.76) would be on completely different scales and no comparison would be meaningful.

The feature-level view ( $\Delta M_f^{\text{ref}}$ ) is what drives the case studies in Section 4.5: it isolates which specific nodes are most destabilized by graph errors. The dataset-level view ( $\Delta M^{\text{ref}}$ ) is what drives the scatter-plot comparisons in Sections 4.2–4.4.

**Sign Disagreement ( $D$ ).** The fraction of (instance, feature) pairs where the attribution flips sign relative to the reference.  $D \in [0, 1]$ ;  $D = 0$  means every attribution points in the same direction as the reference;  $D = 0.33$  means one in three attributions is inverted. Sign flips are qualitatively more damaging than magnitude shifts: a rescaled attribution still tells the same directional story to an affected individual; an inverted one tells the opposite story.

**Three reference configurations.** The same two metrics are computed against three different references, each answering a distinct practical question:

- **vs. Traditional ( $\Delta M_{\text{base}}, D_{\text{base}}$ ):** How much does injecting *any* causal graph change attributions relative to the graph-free baseline? A large value here means the method is structurally sensitive to the graph: it is doing a lot of additional work beyond standard Shapley. Results in Section 4.2.
- **vs. Oracle ( $\Delta M_{\text{oracle}}, D_{\text{oracle}}$ ):** How far does a discovered graph push attributions from what the True (or consensus) DAG would have produced? This measures the cost of graph imperfection directly. Requires a ground-truth graph, so it is only available in controlled settings. Results in Section 4.3.

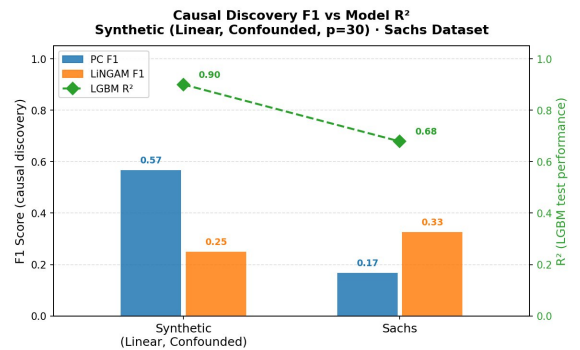
- **PC vs. LiNGAM ( $\Delta M_{\text{disc}}, D_{\text{disc}}$ ):** How much do attributions change when the same method is fed two different discovered graphs from the same data? No ground truth is needed—this is a purely observational stability check available to any practitioner. Results in Section 4.4.

Only the *base* and *disc* metrics are available in a real deployment (oracle requires a known ground-truth graph). The oracle metrics are included here because the controlled experimental setting allows validating the practical diagnostics against a known reference.

## 4 Results and Empirical Analysis

### 4.1 Causal Discovery Performance

On the synthetic track, PC outperforms LiNGAM (F1: 0.567 vs. 0.250): conditional independence tests partially block confounder-induced paths, while LiNGAM’s functional model conflates them with direct paths, adding 81 false-positive edges. On Sachs the ranking reverses (F1: LiNGAM 0.326 vs. PC 0.167): log-transformed proteins retain non-Gaussian marginals that LiNGAM exploits, while PC’s Fisherz test loses discriminative power when the massive sample size makes it hypersensitive to distributional violations [3]. Both algorithms perform substantially worse on the real data despite the larger sample size.



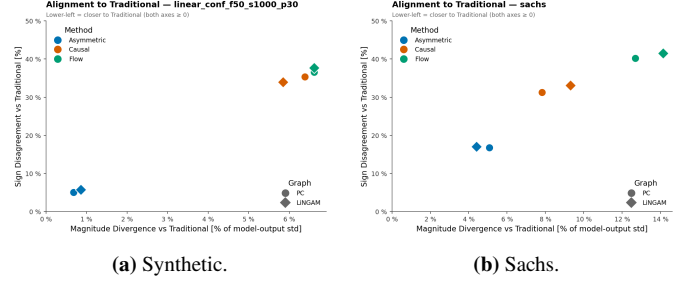
**Figure 2: Causal discovery F1 ( $X \rightarrow X$  edges) versus LightGBM test  $R^2$ , synthetic vs. Sachs.** Bars give PC and LiNGAM F1 on each dataset; the dashed line tracks model  $R^2$ . The discovery-quality ordering of the two algorithms reverses between tracks even as model fit declines from synthetic to real data.

## 4.2 Graph-Free Baseline Deviation

This evaluation block compares each structure-aware method and discovered-graph combination against the Traditional Shapley baseline ( $\Delta M_{\text{base}}$ ,  $D_{\text{base}}$ ) to verify the expected theoretical behaviour of each method and validate that this behaviour holds on real data. The theoretical sensitivity ordering is driven by how deeply each method couples to the graph: ASV uses discovered edges only to restrict the permutation space while keeping the observational value function unchanged, so its baseline deviation depends entirely on how many orderings the graph forbids; CSV is more sensitive because the graph enters the value function itself, driving post-interventional distributions via discovered parent sets; and Flow is the most sensitive because the edges are the players of the game, so any graph error propagates across all downstream paths simultaneously.

Full numeric tables are in Appendix A; scatter plots in Figure 3; feature-level magnitude and sign heatmaps in Appendix A.5–A.6. Two patterns hold across both datasets. First, sign disagreement  $D_{\text{base}}$  is more severe than magnitude divergence  $\Delta M_{\text{base}}$ : injecting a causal graph flips the direction of a larger share of attributions than it inflates their size, because sign inversions arise whenever a change crosses zero and many near-zero attributions sit close to that threshold. Second, the choice between PC and LiNGAM barely matters for the graph-versus-no-graph contrast: both discovered graphs produce nearly identical  $\Delta M_{\text{base}}$  and  $D_{\text{base}}$  for a given Shapley method, so the dominant driver is the method’s structural coupling to any graph, not the specific graph supplied.

**ASV** deviates the least from the Traditional baseline. On synthetic:  $D_{\text{base}} = 5.08\text{--}5.76\%$ ,  $\Delta M_{\text{base}} = 0.68\text{--}0.86\%$ . On Sachs:  $D_{\text{base}} = 16.80\text{--}17.00\%$ ,  $\Delta M_{\text{base}} = 4.41\text{--}5.08\%$ . The larger Sachs numbers reflect its compact 10-node network where each ordering constraint binds more feature pairs. **CSV** changes the sign of roughly a third of all attributions ( $D_{\text{base}} \approx 33\text{--}35\%$  on synthetic,  $\approx 31\text{--}33\%$  on Sachs). **Flow** shows the highest magnitude deviation ( $\Delta M_{\text{base}}$  up to 14.17% of  $\hat{\sigma}$  on Sachs) and the highest sign instability on synthetic ( $D_{\text{base}} = 36.58\text{--}37.68\%$ ).



**Figure 3: Deviation from Traditional Shapley.** ASV clusters lower-left; CSV and Flow upper-right on both axes. Sachs spreads wider.

## 4.3 Oracle Alignment

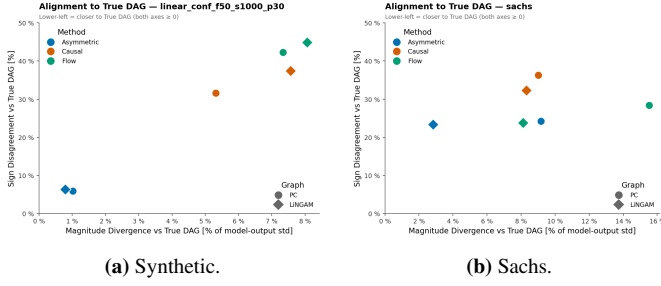
Oracle-alignment measures how closely each (method, discovered graph) recovers the attributions that would be obtained with the True DAG (tables: Appendix A; scatter: Figure 4; feature-level heatmaps: Appendix A.5–A.6). A consistent pattern emerges: the discovery algorithm with better F1 produces smaller oracle deviation on both dimensions. On synthetic, CSV under PC ( $F1 = 0.567$ ) reaches  $D_{\text{oracle}} = 31.60\%$ ,  $\Delta M_{\text{oracle}} = 5.32\%$  vs. 37.46%, 7.57% under LiNGAM ( $F1 = 0.250$ ). On Sachs the ranking reverses: ASV under LiNGAM ( $F1 = 0.326$ ) achieves  $D_{\text{oracle}} = 23.40\%$ ,  $\Delta M_{\text{oracle}} = 2.84\%$  vs. 24.26%, 9.16% under PC ( $F1 = 0.167$ ). Even imperfect graphs help when they are the better of the two available options.

**ASV** achieves near-perfect oracle fidelity on synthetic ( $\Delta M_{\text{oracle}} < 1.1\%$ ,  $D_{\text{oracle}} < 6.3\%$ ) regardless of which discovered graph is supplied. The 81 LiNGAM false-positive edges minimally affect ASV because in a 50-node sparse system the true and discovered graphs share most valid topological orderings. **CSV and Flow** show substantial vulnerability: CSV+LiNGAM yields  $D_{\text{oracle}} = 37.46\%$  and  $\Delta M_{\text{oracle}} = 7.57\%$  on synthetic; Flow shows the largest sign disagreement ( $D_{\text{oracle}} = 42.26\text{--}44.91\%$  on synthetic), meaning close to half of all attributions in all explained features point in the wrong direction relative to the True DAG.

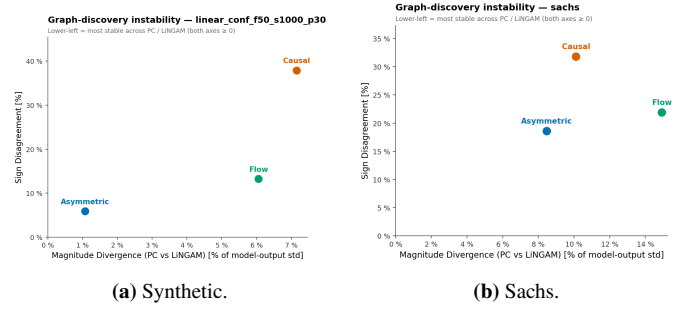
## 4.4 Cross-Discovery Sensitivity

If a practitioner runs PC and LiNGAM on the same data and feeds each graph into the same Shapley method, how much do the explanations change? (Ta-





**Figure 4: Deviation from the oracle DAG.** ASV lower-left under both graphs; CSV and Flow upper-right. PC/LiNGAM markers spread wider on Sachs.



**Figure 5: Cross-discovery instability (PC vs. LiNGAM).** ASV lower-left (stable); CSV upper-right (unstable signs); Flow largest magnitude gap on Sachs.

bles: Appendix A; scatter: Figure 5; feature-level heatmaps: Appendix A.5–A.6.) The answer depends heavily on the Shapley method. On Sachs, Flow’s magnitude divergence between the two graphs ( $\Delta M_{\text{disc}} = 14.92\%$  of  $\hat{\sigma}$ ) is nearly double ASV’s ( $8.46\%$ ), meaning the choice of discovery algorithm alone makes Flow attributions twice as volatile. On the sign axis, switching from ASV to CSV on Sachs determines whether swapping graphs flips  $18.60\%$  or  $31.80\%$  of attribution signs. With this measures, a practitioner now can have a quantitative expectation of how much their attributions might change if they had used a different discovery algorithm, and decide whether that level of instability is acceptable for their use case, and spot possible sources of divergence, by looking at these metrics a more granular level.

ASV shows the lowest cross-discovery sensitivity on both tracks. On synthetic:  $\Delta M_{\text{disc}} = 1.07\%$ ,  $D_{\text{disc}} = 5.93\%$ . On Sachs sensitivity increases ( $\Delta M_{\text{disc}} = 8.46\%$ ,  $D_{\text{disc}} = 18.60\%$ ) as the compact network makes different edge sets impose materially different topological constraints. CSV carries the largest sign instability ( $D_{\text{disc}} = 37.94\%$  synthetic,  $31.80\%$  Sachs): interventional conditioning inverts attribution signs whenever parent sets differ between graphs. Flow carries the largest magnitude instability on Sachs ( $\Delta M_{\text{disc}} = 14.92\%$ ).

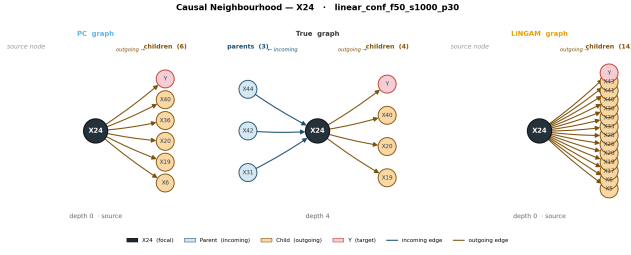
## 4.5 Case Studies: When Graph Errors Matter Most

Three archetypal error mechanisms illustrate the micro-level behaviour observed across both experimental tracks. Each is identified by tracing a large

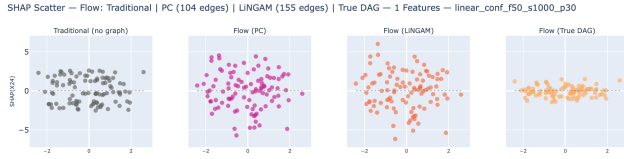
global or feature-level metric back to a specific discovery error.

### 4.5.1 Out-Degree Inflation and Root Cause Overloading

Interventional (CSV) and edge-routing (Shapley Flow) methods over-inflate the attribution of any node that a discovered graph misrepresents as a high-degree root source. The clearest synthetic case is X24, a true direct parent of  $Y$  with 3 incoming and 4 outgoing edges in the True DAG. Both PC and LiNGAM strip its true parents and add spurious children (Figure 6), recasting it as an apparent high-degree root source; this inflates its outgoing-edge credit under Flow and its interventional parent set under CSV. X24’s feature-level  $\Delta M_{\text{base}}$  reaches  $32.5\%$  of  $\hat{\sigma}$ , the highest on the synthetic track, and the SHAP scatter (Figure 7) confirms that both discovered-graph columns diverge substantially from the True-DAG oracle, with inflated and partially sign-flipped values. Any feature producing a  $\Delta M_{\text{base}}$  well above the dataset average deserves scrutiny of its discovered connections before any Flow attribution is used downstream.

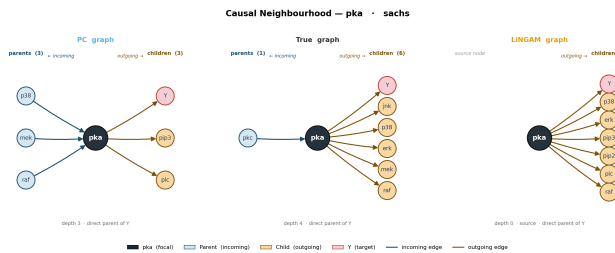


**Figure 6: Causal neighbourhood of X24 (Synthetic).** Both PC and LiNGAM strip true parents and add spurious children (6 and 14 out-edges respectively, 0 in-edges), converting a mid-graph node into an apparent root source. Blue = incoming, amber = outgoing.



**Figure 7: Shapley Flow SHAP scatter for X24 (Synthetic).** Traditional shows a stable attribution pattern; both discovered-graph columns diverge substantially from the True-DAG oracle—the instance-level signature of out-degree inflation ( $\Delta M_{\text{base}} = 32.5\%$  of  $\hat{\sigma}$ ).

The same mechanism appears on Sachs through protein pka, a near-root source (1 incoming, 6 outgoing edges in the consensus DAG). LiNGAM preserves this structure accurately, keeping pka as a near-pure root (ASV  $\Delta M_{\text{oracle}} = 6.4\%$  of  $\hat{\sigma}$ ). PC reverses several of its outgoing edges into a 3-in/3-out configuration, and that mis-orientation raises pka’s ASV  $\Delta M_{\text{oracle}}$  to 25.0% (Figure 8). A high  $\Delta M_{\text{base}}$  (17.3% of  $\hat{\sigma}$ ) combined with a large  $\Delta M_{\text{disc}}$  forms the two-signal alarm: pka’s discovered connections are both influential and contested across algorithms.

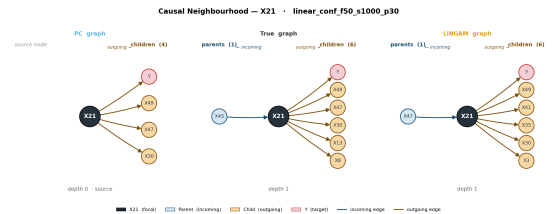


**Figure 8: Causal neighbourhood of pka (Sachs).** PC gives pka 3 incoming and 3 outgoing edges, reversing several true outgoing links. LiNGAM keeps it a near-root with 7 outgoing and 0 incoming edges, much closer to the consensus structure.

## 4.5.2 Directed Edge Inversion and Causal Credit Transfer

A single edge orientation error by one discovery algorithm can produce a sharply divergent attribution. In the True DAG the edge runs  $X_{21} \rightarrow X_{47}$ :  $X_{21}$  is a causal ancestor of  $X_{47}$ , which is a dominant direct parent of  $Y$ . LiNGAM recovers this orientation correctly; PC reverses it to  $X_{47} \rightarrow X_{21}$ , placing  $X_{47}$  upstream of  $X_{21}$  (Figure 9).

Under LiNGAM, both CSV and Flow correctly treat  $X_{21}$  as an upstream node, assigning indirect credit for its causal influence through  $X_{47}$ . Under PC the reversed edge injects  $X_{47}$  into  $X_{21}$ ’s interventional parent set (CSV) and routes edge credit from  $X_{47}$  toward  $X_{21}$  (Flow), suppressing  $X_{47}$ ’s attribution and inflating  $X_{21}$ ’s. The SHAP scatters (Figure 10) confirm this: both CSV+LiNGAM and Flow+LiNGAM track the True-DAG oracle closely, while the PC variants diverge on both features. Feature-level  $\Delta M_{\text{disc}}$  for CSV reaches 29.8% of  $\hat{\sigma}$  on  $X_{47}$ —the two-signal alarm indicating that a real edge’s *orientation*, not its existence, must be resolved before using any structure-aware Shapley method.



**Figure 9: Causal neighbourhood of X21 (Synthetic).** In the True DAG and under LiNGAM the edge runs  $X_{21} \rightarrow X_{47}$ . PC reverses it to  $X_{47} \rightarrow X_{21}$ , misdirecting attribution credit in both CSV and Flow.

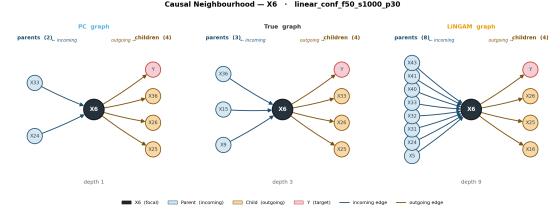




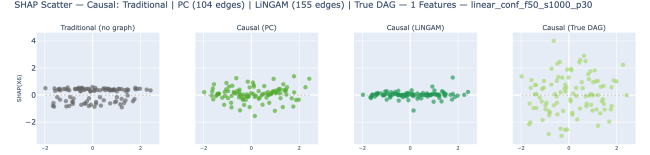
**Figure 10: SHAP scatter for X21 and X47 (Synthetic).** *Top* (Causal Shapley): under LiNGAM both features track the oracle; under PC, X47 is suppressed and X21 acquires inflated credit. *Bottom* (Shapley Flow): the same orientation-driven credit transfer appears; Flow+LiNGAM and Flow+True agree closely while Flow+PC diverges on both features.

### 4.5.3 Confounded Subordination and Boundary Isolation

Synthetic feature X6 is a true intermediate node with 3 parents and 4 children. LiNGAM inflates its in-degree to 8 with spurious incoming links; under CSV these false parents generate an overly constrained post-interventional distribution, yielding a feature-level  $\Delta M_{\text{oracle}} = 28.1\%$  of  $\hat{\sigma}$ —among the highest in the synthetic confounded track (Figure 11). The SHAP scatter (Figure 12) shows the LiNGAM column diverging sharply from both the Traditional baseline and the True-DAG oracle while PC stays close—a direct consequence of LiNGAM inflating X6’s in-degree and contaminating its interventional conditioning set. This makes the precision and recall of the discovery step a direct, visible factor in CSV’s reliability.

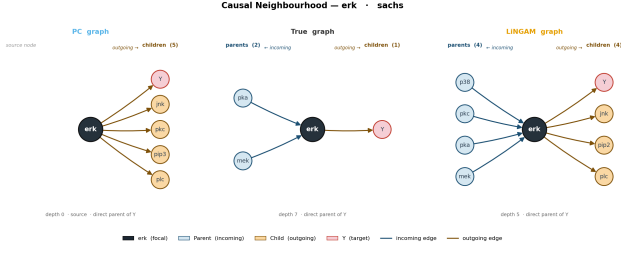


**Figure 11: Causal neighbourhood of X6 (Synthetic).** X6 has 3 incoming and 4 outgoing edges in the True DAG. LiNGAM inflates its in-degree to 8, contaminating its interventional conditioning set under CSV.

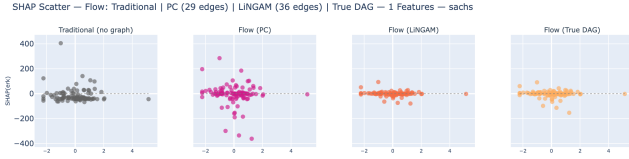


**Figure 12: Causal Shapley SHAP scatter for X6 (Synthetic).** The LiNGAM column diverges sharply from both the Traditional baseline and the True-DAG oracle while PC stays close—a direct consequence of LiNGAM’s in-degree inflation from 3 to 8.

On Sachs, protein erk is a downstream node receiving inputs from pka and mek in the consensus DAG. PC removes its parent links and isolates it as a root source with several outgoing edges (Figure 13), giving erk a disproportionately large Flow attribution ( $\Delta M_{\text{oracle}} = 49.2\%$  of  $\hat{\sigma}$  for Flow+PC, the largest single deviation in the real-data experiment). The SHAP scatter (Figure 14) shows all three graph-aware variants producing materially different attribution patterns: Flow+PC inflates erk by treating it as a source; Flow+LiNGAM produces a distinct distortion; only Flow+True DAG tracks the correct downstream credit distribution. Across all three metrics— $\Delta M_{\text{base}}$ ,  $\Delta M_{\text{oracle}}$ , and  $\Delta M_{\text{disc}}$ —erk is simultaneously far from the graph-free baseline, far from the oracle, and far from itself under a different algorithm. This triple divergence is the framework’s strongest warning signal.



**Figure 13: Causal neighbourhood of erk (Sachs).** PC removes its true parent links (pka, mek) and isolates erk as a boundary source—driving Flow+PC’s  $\Delta M_{\text{oracle}} = 49.2\%$  of  $\hat{\sigma}$ .



**Figure 14: Shapley Flow SHAP scatter for erk (Sachs).** All three graph-aware variants diverge from each other and from Traditional. Flow+PC inflates erk’s attributions by treating it as an isolated source node; only Flow+True DAG tracks the correct downstream credit distribution.

## 5 Conclusion

### 5.1 Summary of Findings

This work built an experimental pipeline to measure what happens to feature attributions when you plug an automatically discovered causal graph into a structure-aware Shapley method. The three Shapley methods fall into completely different sensitivity regimes. ASV is stable under graph errors, specially in sparse high-dimensional datasets: on the 50-feature synthetic data, swapping the True DAG for a discovered graph moves ASV’s attributions by less than 1.1% of  $\hat{\sigma}$  and flips fewer than 7% of attribution signs. This stability comes from ASV’s core logic—it only restricts the permutation space, leaving the observational value function untouched, so most topological orderings remain compatible across the true and discovered graphs. CSV and Flow’s core logic instead injects the graph into the conditioning set or into the players themselves, so every false edge contaminates an attribution directly. Even though, the experimental pipeline allows spotting features that have structural divergences shifting the explanations badly, which

would allow practitioners to spot improvement points even in ASV before committing to any structure-aware attribution.

The two discovery algorithms swap rankings across datasets (PC better on synthetic, LiNGAM better on Sachs) and neither reaches high accuracy on either. The real question is not which algorithm is generally better, but how much the chosen Shapley method amplifies the errors that algorithm makes. A method sensitive to graph quality, like CSV or Flow, inherits the weaknesses of whatever discovery algorithm is used; ASV absorbs them. This shifts the practical decision: rather than finding the best discovery algorithm, it is more productive to first check how sensitive the Shapley method is to graph errors on the specific data at hand, and only invest in graph improvement where sensitivity is high.

The evaluation metrics reveal things you cannot see by looking at attributions alone. A high  $\Delta M_{\text{base}}$  on a node flags that the injected graph is doing a lot of work there (X24: 32.5%, X6: 28.1%, pka: 17.3%). When  $\Delta M_{\text{base}}$  and  $\Delta M_{\text{disc}}$  are both high on the same node, the graph structure for that feature is both influential and contested: the two-signal alarm. erk on Sachs adds a third signal ( $\Delta M_{\text{oracle}}$  also high), the clearest warning the framework can emit.

### 5.2 Practical Recommendations

The most important contribution is the evaluation framework itself. Before this work, a practitioner adding a discovered graph to a Shapley computation had no systematic way to assess what that decision would cost in attribution quality. The six metrics at both dataset and feature level give a structured vocabulary for this assessment. In most real-world scenarios only the base and disc metrics are available; those are the ones the framework is really built around. The practical ordering is clear:

- **ASV is the recommended default.** It delivers oracle-level alignment with minimal deviation from the Traditional baseline, works well even with poor discovery results (Sachs PC: F1 = 0.167), and adds no computational cost above standard Monte Carlo SHAP.
- **CSV and Flow are useful only when graph**

**quality is high**—when domain experts have validated the structure, or when the cross-discovery stability metrics confirm the graph is stable. CSV+LiNGAM on confounded synthetic produces  $D_{\text{oracle}} = 37.5\%$  and  $D_{\text{disc}} = 37.9\%$ : attributions differ in sign from the oracle more than a third of the time and disagree with themselves across graph sources at the same rate. That level of instability makes explanations inconsistent across audit runs and difficult to defend in practice.

### 5.3 Limitations and Future Work

The analysis covers a single synthetic functional form (linear confounded) and one real dataset; nonlinear and mixed functional forms, different graph densities, and larger sample sizes remain to be tested. All Shapley computations use  $T = 100$  Monte Carlo samples. Future work should extend to nonlinear Shapley variants, explore ensemble approaches averaging attributions across multiple discovered graphs to reduce cross-discovery instability, and investigate whether active learning strategies could use the feature-level  $\Delta M_{\text{disc}}$  signal to guide targeted graph refinement starting from the nodes the framework identifies as most contested.

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## A Detailed Results Tables

### A.1 Baseline Deviation (vs. Traditional Shapley)

**Table 3: Baseline Deviation – Linear-Conf Synthetic Dataset.**

| Method     | Graph  | $D_{\text{base}}$ | $\Delta M_{\text{base}}$ |
|------------|--------|-------------------|--------------------------|
| Asymmetric | PC     | 5.08%             | 0.68%                    |
| Asymmetric | LiNGAM | 5.76%             | 0.86%                    |
| Causal     | PC     | 35.30%            | 6.39%                    |
| Causal     | LiNGAM | 33.92%            | 5.84%                    |
| Flow       | PC     | 36.58%            | 6.62%                    |
| Flow       | LiNGAM | 37.68%            | 6.61%                    |

**Table 4: Baseline Deviation – Sachs Cell Signaling Dataset.**

| Method     | Graph  | $D_{\text{base}}$ | $\Delta M_{\text{base}}$ |
|------------|--------|-------------------|--------------------------|
| Asymmetric | PC     | 16.80%            | 5.08%                    |
| Asymmetric | LiNGAM | 17.00%            | 4.41%                    |
| Causal     | PC     | 31.30%            | 7.83%                    |
| Causal     | LiNGAM | 33.10%            | 9.31%                    |
| Flow       | PC     | 40.20%            | 12.68%                   |
| Flow       | LiNGAM | 41.50%            | 14.17%                   |

### A.2 Oracle Alignment (vs. True / Consensus DAG)

**Table 5: Oracle Alignment – Linear-Conf Synthetic Dataset.**  $D_{\text{oracle}}$  and  $\Delta M_{\text{oracle}}$  in % of  $\hat{\sigma}$ .

| Method     | Graph  | $D_{\text{oracle}}$ | $\Delta M_{\text{oracle}}$ |
|------------|--------|---------------------|----------------------------|
| Asymmetric | PC     | 5.88%               | 1.02%                      |
| Asymmetric | LiNGAM | 6.30%               | 0.79%                      |
| Causal     | PC     | 31.60%              | 5.32%                      |
| Causal     | LiNGAM | 37.46%              | 7.57%                      |
| Flow       | PC     | 42.26%              | 7.34%                      |
| Flow       | LiNGAM | 44.91%              | 8.06%                      |

**Table 6: Oracle Alignment – Sachs Cell Signaling Dataset (oracle = consensus DAG).**  $D_{\text{oracle}}$  and  $\Delta M_{\text{oracle}}$  in % of  $\hat{\sigma}$ .

| Method     | Graph  | $D_{\text{oracle}}$ | $\Delta M_{\text{oracle}}$ |
|------------|--------|---------------------|----------------------------|
| Asymmetric | PC     | 24.26%              | 9.16%                      |
| Asymmetric | LiNGAM | 23.40%              | 2.84%                      |
| Causal     | PC     | 36.30%              | 9.01%                      |
| Causal     | LiNGAM | 32.30%              | 8.31%                      |
| Flow       | PC     | 28.36%              | 15.52%                     |
| Flow       | LiNGAM | 23.81%              | 8.12%                      |

### A.3 Cross-Discovery Sensitivity (PC vs. LiNGAM)

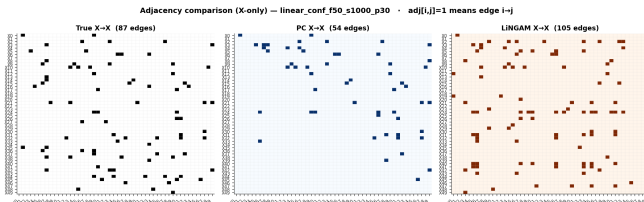
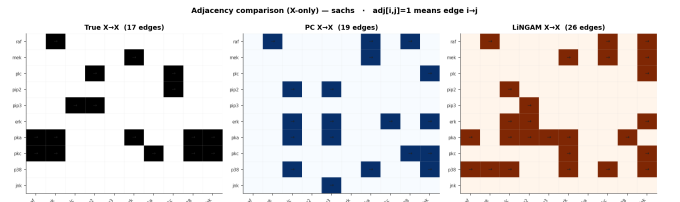
**Table 7: PC vs. LiNGAM Sensitivity – Linear-Conf Synthetic Dataset.**

| Method     | $\Delta M_{\text{disc}}$ | $D_{\text{disc}}$ |
|------------|--------------------------|-------------------|
| Asymmetric | 1.07%                    | 5.93%             |
| Causal     | 7.15%                    | 37.94%            |
| Flow       | 6.06%                    | 13.27%            |

**Table 8: PC vs. LiNGAM Sensitivity – Sachs Cell Signaling Dataset.**

| Method     | $\Delta M_{\text{disc}}$ | $D_{\text{disc}}$ |
|------------|--------------------------|-------------------|
| Asymmetric | 8.46%                    | 18.60%            |
| Causal     | 10.11%                   | 31.80%            |
| Flow       | 14.92%                   | 21.88%            |

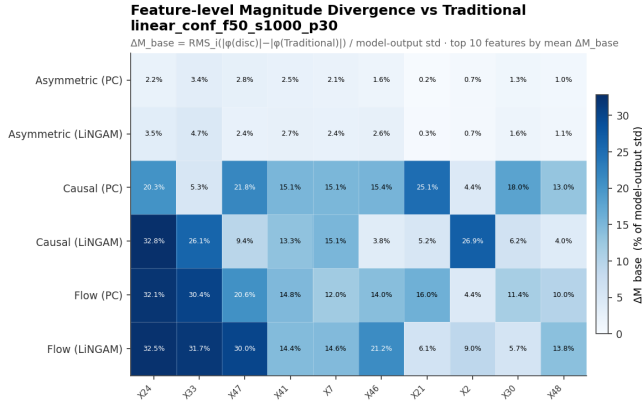
### A.4 Discovery Adjacency Matrices

**(a) Synthetic:** True DAG (87 edges), PC (54), LiNGAM (105). LiNGAM scatters spurious edges across the matrix.**(b) Sachs:** consensus DAG (17 edges), PC (19), LiNGAM (26). Both algorithms recover only a fraction of true links.**Figure 15:  $X \rightarrow X$  adjacency matrices.**  $\text{adj}[i, j] = 1$  denotes  $i \rightarrow j$ . The discovery-quality ordering reverses between datasets.

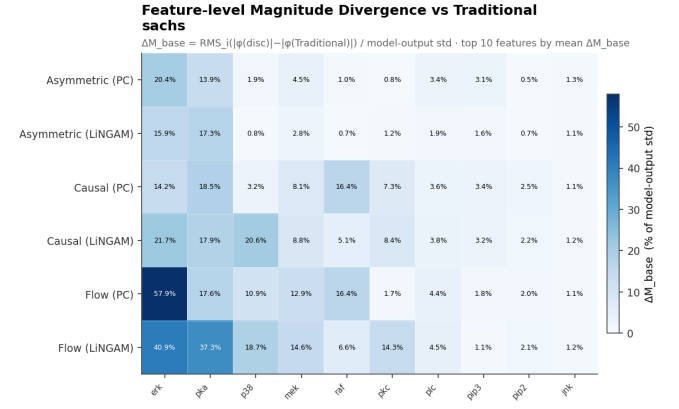
### A.5 Feature-Level Magnitude Heatmaps ( $\Delta M$ )

Each heatmap shows  $\Delta M_f^{\text{ref}}$  for the top 10 features by mean divergence. Rows are method-graph combinations; columns are features. Darker cells indicate larger attribution magnitude shifts. The same top features dominate all three comparisons.



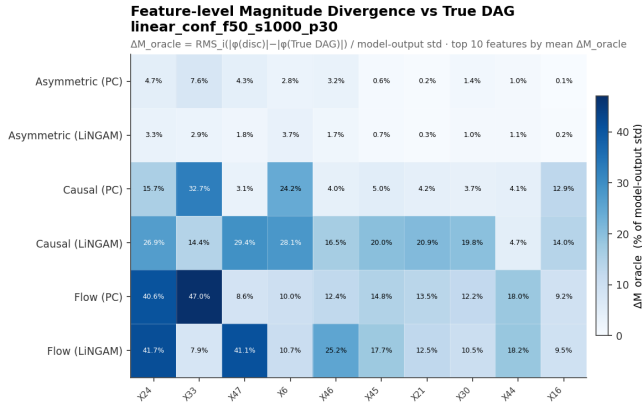


(a) Synthetic. Asymmetric rows near-zero throughout; Causal and Flow concentrate on X24, X33, X47.

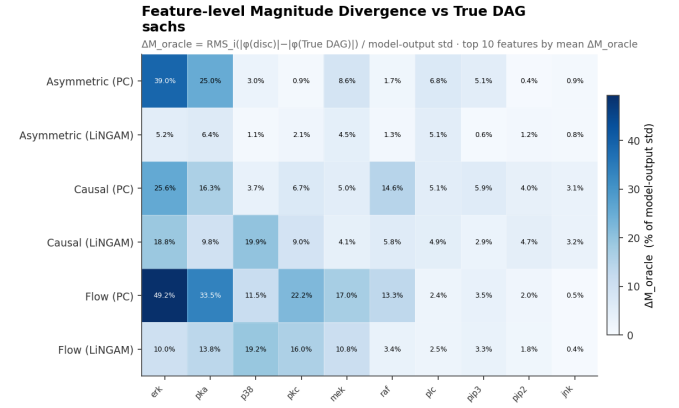


(b) Sachs. Deviation concentrates on erk and pka; Flow+PC produces the largest single shift (erk, 57.9% of  $\hat{\sigma}$ ).

**Figure 16: Feature-level  $\Delta M_{base}$  vs. Traditional Shapley.**

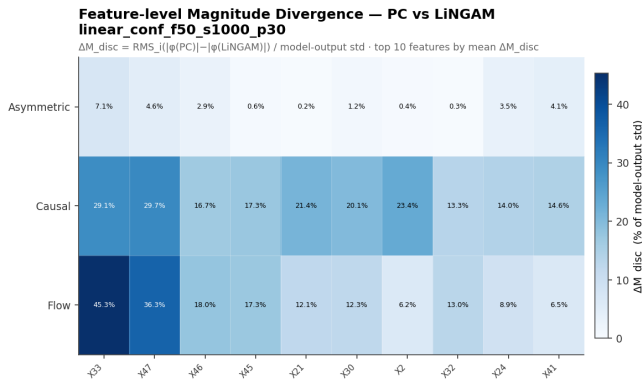


(a) Synthetic. Asymmetric rows near-zero; Causal and Flow concentrate oracle deviations on X24, X33, X47, X6.

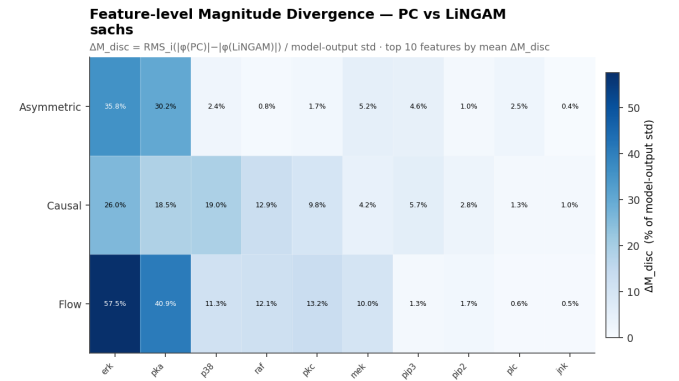


(b) Sachs. Flow+PC produces the largest single deviation on erk (49.2% of  $\hat{\sigma}$ ).

**Figure 17: Feature-level  $\Delta M_{oracle}$  vs. the oracle DAG.**



(a) Synthetic. Asymmetric near-zero; Causal and Flow concentrate on X33 and X47.

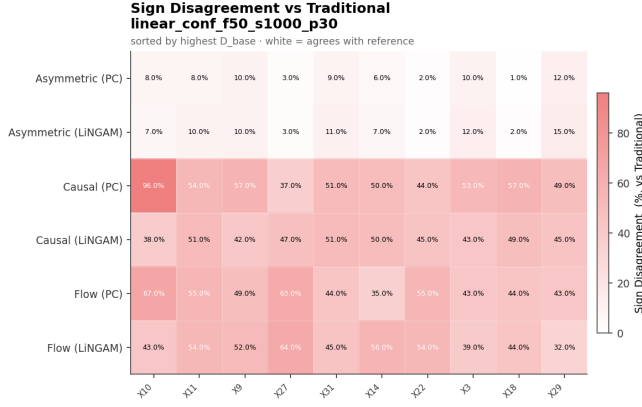


(b) Sachs. Sensitivity concentrates on erk and pka; Flow reaches 57.5% of  $\hat{\sigma}$  on erk.

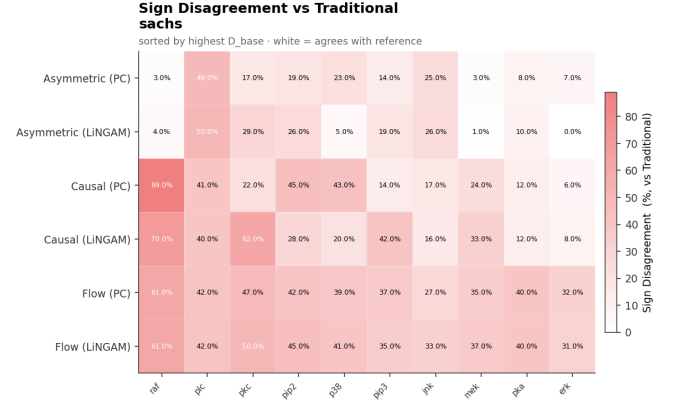
**Figure 18: Feature-level  $\Delta M_{disc}$  (PC vs. LiNGAM).**

## A.6 Feature-Level Sign Disagreement Heatmaps ( $D$ )

Each heatmap shows  $D_f^{\text{ref}}$  for the top 10 features by mean sign disagreement. Sign flips are qualitatively more damaging than magnitude shifts (they invert the story told to the end user), and they tend to spread across more features than magnitude changes because many near-zero attributions sit close to the sign boundary.

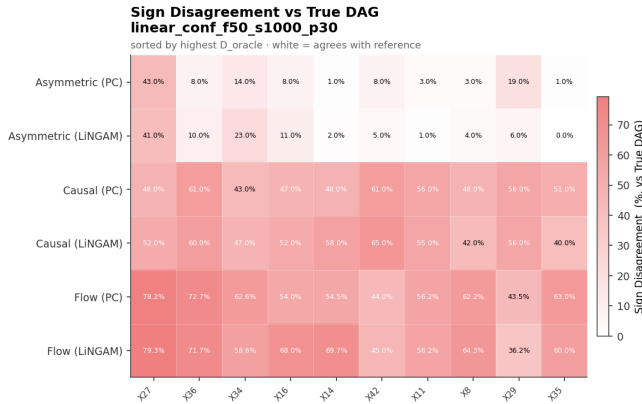


(a) Synthetic. Asymmetric rows near-zero; Causal and Flow concentrate sign flips on X24, X33, X47.

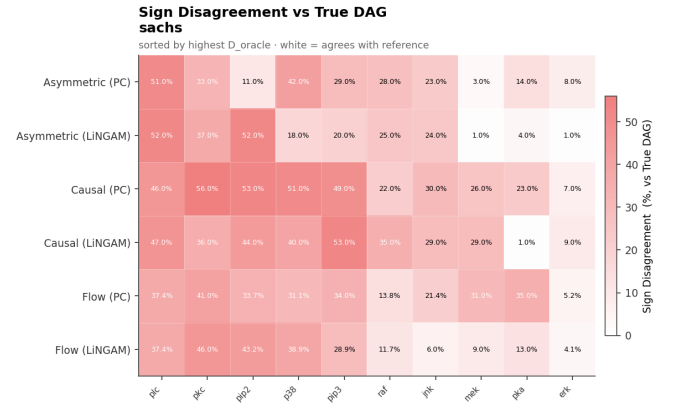


(b) Sachs. Sign disagreement concentrates on raf, plc, pkc; Causal+LiNGAM produces the highest per-feature flip rate vs. the baseline.

Figure 19: Feature-level  $D_{\text{base}}$  vs. Traditional Shapley.

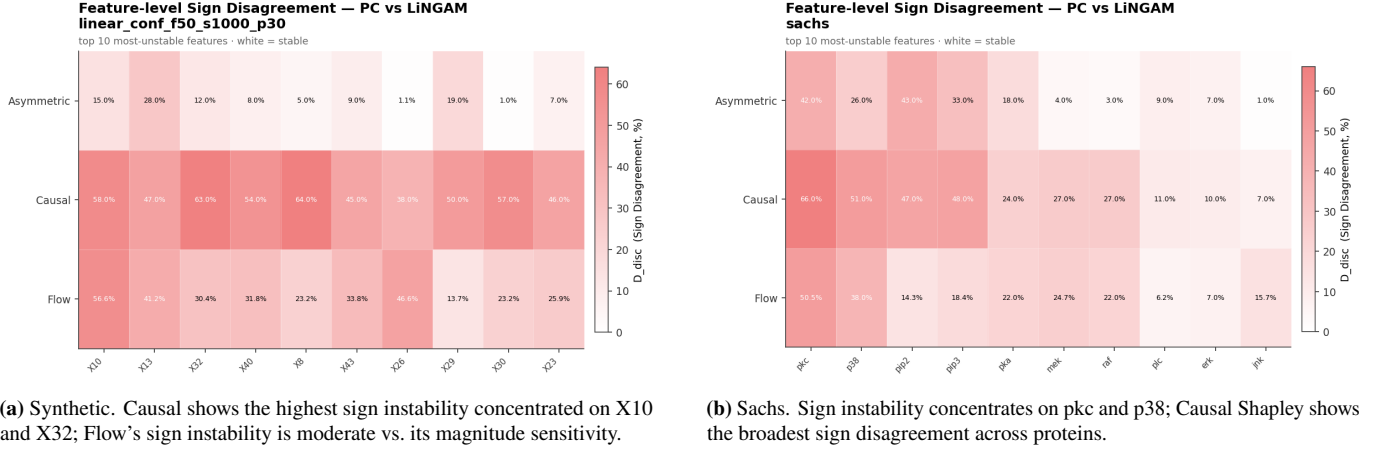


(a) Synthetic. Causal and Flow concentrate sign disagreement on X24, X33, X47, X6 — the same features that carry the oracle magnitude deviations.



(b) Sachs. Sign disagreement vs. oracle concentrates on erk and pkc; Flow+PC has the highest per-feature oracle sign-flip rate on erk.

Figure 20: Feature-level  $D_{\text{oracle}}$  vs. the oracle DAG.

Figure 21: Feature-level  $D_{\text{disc}}$  (PC vs. LiNGAM).

## B Algorithm Pseudocode

The four main algorithms and five helper subroutines are listed below. They implement the methods described in Section 2.

### B.1 Main Methods

---

#### Algorithm 1 Traditional Shapley (Monte Carlo)

---

**INPUT:** instance  $x$ , model  $f$ , background data  $D$ , permutations  $T$

**OUTPUT:** attribution vector  $\phi \in \mathbb{R}^n$

```

1:  $\phi \leftarrow \mathbf{0}_n$ ;  $v_0 \leftarrow \text{mean}(f(D))$ 
2: for  $t = 1 \dots T$  do
3:    $\pi \leftarrow \text{RANDOM\_PERMUTATION}(1 \dots n)$ 
4:    $S \leftarrow \emptyset$ ;  $v_{\text{prev}} \leftarrow v_0$ 
5:   for  $i$  in  $\pi$  do
6:      $S \leftarrow S \cup \{i\}$ ;  $v_{\text{curr}} \leftarrow \text{COALITION\_VALUE}(x, S, f, D)$ 
7:      $\phi[i] \leftarrow \phi[i] + (v_{\text{curr}} - v_{\text{prev}})$ ;  $v_{\text{prev}} \leftarrow v_{\text{curr}}$ 
8:   end for
9: end for return  $\phi / T$ 

```

---

**Algorithm 2** Asymmetric Shapley**INPUT:** instance  $x$ , model  $f$ , background  $D$ , feature DAG  $G$ , permutations  $T$ **OUTPUT:** attribution vector  $\phi \in \mathbb{R}^n$ **PRECOMPUTE:** children[], in\_degree[] from the  $X$ -only edges of  $G$ 

```

1:  $\phi \leftarrow \mathbf{0}_n$ ;  $v_0 \leftarrow \text{mean}(f(D))$ 
2: for  $t = 1 \dots T$  do
3:    $\pi \leftarrow \text{SAMPLE\_TOPOLOGICAL\_ORDERING}()$ 
4:    $S \leftarrow \emptyset$ ;  $v_{\text{prev}} \leftarrow v_0$ 
5:   for  $i$  in  $\pi$  do
6:      $S \leftarrow S \cup \{i\}$ ;  $v_{\text{curr}} \leftarrow \text{COALITION\_VALUE}(x, S, f, D)$ 
7:      $\phi[i] \leftarrow \phi[i] + (v_{\text{curr}} - v_{\text{prev}})$ ;  $v_{\text{prev}} \leftarrow v_{\text{curr}}$ 
8:   end for
9: end for return  $\phi / T$ 

```

**Algorithm 3** Causal Shapley (post-interventional)**INPUT:** instance  $x$ , model  $f$ , background  $D$ , adjacency  $A$  ( $X$  only), confounder list, outer permutations  $T$ , inner samples  $M$ **OUTPUT:** attribution vector  $\phi \in \mathbb{R}^n$ **PRECOMPUTE:** components, confounded[], parents[]  $\leftarrow \text{BUILD\_COMPONENTS}(A, \text{confounder list})$ 

```

1:  $\mu \leftarrow \text{mean}(D)$ ;  $\Sigma \leftarrow \text{cov}(D) + \varepsilon I$ 
2:  $\phi \leftarrow \mathbf{0}_n$ ;  $v_0 \leftarrow \text{mean}(f(D))$ 
3: for  $t = 1 \dots T$  do
4:    $\pi \leftarrow \text{SAMPLE\_COMPONENT\_TOPO\_ORDERING}()$ , expand to features
5:    $S \leftarrow \emptyset$ ;  $v_{\text{prev}} \leftarrow v_0$ 
6:   for  $i$  in  $\pi$  do
7:      $S \leftarrow S \cup \{i\}$ 
8:      $v_{\text{curr}} \leftarrow \frac{1}{M} \sum_{m=1}^M f(\text{POST\_INTERVENTIONAL\_SAMPLE}(x, S, \dots))$ 
9:      $\phi[i] \leftarrow \phi[i] + (v_{\text{curr}} - v_{\text{prev}})$ ;  $v_{\text{prev}} \leftarrow v_{\text{curr}}$ 
10:  end for
11: end for return  $\phi / T$ 

```

**Algorithm 4** Shapley Flow (uniform edge-permutation)

**INPUT:** foreground instance  $x_{fg}$ , background row  $x_{bg}$ , graph adjacency ( $n + 1$  nodes incl.  $Y$ ), target index  $Y$ , trials  $T$

**OUTPUT:** node attribution vector  $\phi \in \mathbb{R}^n$  ( $Y$  excluded)

```

1:  $E \leftarrow$  list of all directed edges;  $ec \leftarrow \{e : 0.0 \mid e \in E\}$ 
2: for  $t = 1 \dots T$  do
3:    $perm \leftarrow \text{RANDOM\_PERMUTATION}(E)$ ;  $active \leftarrow []$ 
4:    $v_{prev} \leftarrow \text{SYSTEM\_VALUE}(\emptyset, x_{fg}, x_{bg})$ 
5:   for  $e$  in  $perm$  do
6:      $active.append(e)$ ;  $v_{curr} \leftarrow \text{SYSTEM\_VALUE}(active, x_{fg}, x_{bg})$ 
7:      $ec[e] \leftarrow ec[e] + (v_{curr} - v_{prev})$ ;  $v_{prev} \leftarrow v_{curr}$ 
8:   end for
9: end for
10:  $\phi \leftarrow \mathbf{0}_n$ 
11: for each edge  $(u \rightarrow v) \in E$  with  $u \neq Y$  do
12:    $\phi[u] \leftarrow \phi[u] + ec[(u \rightarrow v)] / T$ 
13: end for return  $\phi$ 

```

**B.2 Subroutines****Algorithm 5** COALITION\_VALUE —  $v(S) = \mathbb{E}[f(X) \mid X_S = x_S]$ 

```

1: function COALITION_VALUE( $x, S, f, D$ )
2:    $samples \leftarrow \text{copy}(D)$ 
3:   for each feature  $j \in S$  do
4:      $samples[:, j] \leftarrow x[j]$ 
5:   end for return  $\text{mean}(f(samples))$ 
6: end function

```

**Algorithm 6** SAMPLE\_TOPOLOGICAL\_ORDERING — uniform random linear extension (Kahn)

**PRECOMPUTE:** (once):  $children[], in\_degree[]$  from the DAG ( $Y$  excluded).

If the feature DAG contains a cycle, disable constraints  $\Rightarrow$  reduces to Traditional Shapley.

```

1: function SAMPLE_TOPOLOGICAL_ORDERING( $children, in\_degree$ )
2:    $deg \leftarrow \text{copy}(in\_degree)$ ;  $ready \leftarrow [i : deg[i] = 0]$ ;  $order \leftarrow []$ 
3:   while  $ready \neq \emptyset$  do
4:      $node \leftarrow \text{uniform pick from ready}$ ;  $\text{swap-remove from ready}$ 
5:      $order.append(node)$ 
6:     for  $c \in children[node]$  do
7:        $deg[c] -= 1$ ; if  $deg[c] = 0$ :  $ready.append(c)$ 
8:     end for
9:   end while return  $order$ 
10: end function

```

**Algorithm 7** POST\_INTERVENTIONAL\_SAMPLE — draw from  $P(X \mid \text{do}(X_S = x_S))$ 


---

```

1: function POST_INTERVENTIONAL_SAMPLE( $x, S, \text{components}, \text{confounded}[], \text{parents}[], \mu, \Sigma$ )
2:    $\text{sample} \leftarrow \mathbf{0}_n$ ; for  $j \in S$ :  $\text{sample}[j] \leftarrow x[j]$ 
3:   for component  $C$  in fixed topological order do
4:      $\text{missing} \leftarrow C \setminus S$ ; if  $\text{missing} = \emptyset$ : continue
5:     if  $\text{confounded}[C]$  then
6:       for  $j \in \text{missing}$  do
7:          $\text{sample}[j] \leftarrow \text{GAUSSIAN\_CONDITIONAL}(\{j\}, \text{parents}[C], \mu, \Sigma)$ 
8:       end for
9:     else
10:       $\text{sample}[\text{missing}] \leftarrow \text{GAUSSIAN\_CONDITIONAL}(\text{missing}, \text{parents}[C] \cup (C \cap S), \mu, \Sigma)$ 
11:    end if
12:  end for return  $\text{sample}$ 
13: end function

```

---

**Algorithm 8** GAUSSIAN\_CONDITIONAL — closed-form conditional Gaussian draw

---

```

1: function GAUSSIAN_CONDITIONAL( $\text{target } A, \text{cond } B, \text{vals } b, \mu, \Sigma$ )
2:   if  $B = \emptyset$  then return draw from  $\mathcal{N}(\mu_A, \Sigma_{AA})$ 
3:   end if
4:    $\Sigma_{BB}^- \leftarrow \text{PSEUDO\_INVERSE}(\Sigma_{BB})$ 
5:    $\mu_c \leftarrow \mu_A + \Sigma_{AB} \Sigma_{BB}^- (b - \mu_B)$ 
6:    $\Sigma_c \leftarrow \Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^- \Sigma_{BA}$ ; symmetrise; nudge eigenvalues  $\geq \varepsilon$  return draw from  $\mathcal{N}(\mu_c, \Sigma_c)$ 
7: end function

```

---

**Algorithm 9** SYSTEM\_VALUE — binary foreground/background activation (Shapley Flow)

---

```

1: function SYSTEM_VALUE( $\text{active\_edges}, x_{\text{fg}}, x_{\text{bg}}$ )
2:   for each node  $i$  in the graph do
3:     if ( $i$  is a source and has an active outgoing edge) or ( $i$  has an active incoming edge) or (edge  $i \rightarrow Y$  is active) then
4:        $\text{val}[i] \leftarrow x_{\text{fg}}[i]$ 
5:     else
6:        $\text{val}[i] \leftarrow x_{\text{bg}}[i]$ 
7:     end if
8:   end for
9:   drop  $Y$  from  $\text{val}$  return  $f(\text{val})$ 
10: end function

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